Samuel Baumgartner

Switzerland | sbaumgartner@ethz.ch | samuelbaumgartner.ch

Profile

Third-year Electrical Engineering student at ETH Zürich with a GPA of 5.7 (top 5%). Focus areas: robotics, embedded systems and control theory. Experienced in building full-stack applications, autonomous systems, and production-grade infrastructure. Planning to pursue a Master's in Robotics at ETH.

Education

Gymnasium Oberwil – Matura (Mathematics & Physics) 2016 – 2022

Swiss Armed Forces – Sergeant 2022 – 2023
- Technical responsibility for digital and computer systems
- Experience in secure, mission-critical infrastructure operation

ETH Zürich – BSc Electrical Engineering (GPA: 5.7/6.0) 2023 – Present
- Focus: Control Systems, Robotics, Embedded Systems
- Teaching Assistant at ETH
- Target degree: MSc Robotics, Systems and Control

Technical Skills

Programming & Scientific Computing: C++, Python, NumPy, SciPy, Matplotlib
Embedded & Hardware: Arduino, Raspberry Pi, ESP8266, motor control, sensor integration, wireless systems
Machine Learning & AI: TensorFlow, PyTorch, OpenCV; deep learning, computer vision
Robotics & Control: PID control, RRT / RRT* / RRT* Smart, real-time software
Systems & DevOps: Linux, Git, Docker, Docker Compose, shell scripting, system administration
3D Modeling: Blender, ZBrush, Substance Painter, PrusaSlicer

Hackathon Projects

Guidely: AI Desktop Guidance Assistant – Swiss AI Hack 2025 (2nd Place) Nov 2025
- Built core AI pipeline and desktop overlay in < 24 hours
- Stack: Electron, React, TypeScript, OpenAI, Python; 2nd of 40 teams (CHF 3'000)

UrbanLens – NASA Space Apps Challenge (3rd Place) Oct 2025
- Developed Python API and transportation data layer
- Stack: Next.js, FastAPI, GeoPandas, Rasterio, scikit-learn, Vercel

Projects

Ball on a Plate – Motion Planning Algorithms 2025
- Implemented RRT, RRT*, and RRT* Smart in Python with full visualisation for position on control surface

Remote Controlled Robotic Ball 2025
- Designed a spherical robot controllable via smartphone interface, including a internal mass-shifting mechanism
- Developed embedded firmware and created full 3D mechanical models and components in Blender

Experience

Teaching Assistant, ETH Zürich 2024 – Present
- Leading weekly exercise sessions for Electrical Engineering courses
- Supporting students in electrical circuit analysis and magnetic fields

Interests

Robotics, control engineering, embedded systems, SaaS development, fitness, Japanese language study, beekeeping